MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

BICT 4703: NETWORK ADMINISTRATION AND INFORMATION SECURITY

END OF SEMESTER EXAMINATION

DATE: September,2021 **TIME ALLOWED: 2 HRS 30 MIN**

INSTRUCTIONS

i. There are **5** questions in this paper. **ANSWER ALL.**

i. There are **TWO** pages. **PLEASE CHECK**

iii. Marks are indicated for each question.

iii. Please begin each answer on a fresh page and indicate the question number on each page

iii. Write your answers clearly.

Question 1

a. Define the following concepts:

i. Wired Equivalent Privacy Protocol (WEP). **[2 Marks]**

ii. Authentication. **[2 Marks]**

iii. Encryption. **[2 Marks]**

b. Explain the **three** concepts of the CIA triad and for each give an example of how you would achieve this in securing your network. **[6 Marks]**

c. Discuss the **three** components of an intrusion detection system. **[6 Marks]**

d. Explain **one** challenge of an intrusion detection system. **[2 Marks]**

Question 2

a) Explain how a Virtual Private Network(VPN) works. **[4 Marks]**

b) Discuss the concept of One Time Passwords (OTP) and how they are used to authenticate users accessing critical applications. In your explanation include **two** advantages of incorporating OTP into financial applications. **[6 Marks]**

c) Between a software and a hardware token, which one would you implement? Support your answer. **[4 Marks]**

d) What is a computer Virus? **[2 Marks]**

e) Mention any 4 indicators of a computer being infected by a virus. **[4 Marks]**

Question 3

The year is 2021 and the company 'Fictitious Corp.' starts its business day at 8 a.m. as usual. At around 11 a.m., one of the IT managers hears a strange sound coming from a nearby area. Just after hearing the noise, his phone rings and he answers it. After finishing the call, he realizes that the workstation is totally paralyzed and reads a message on the screen saying all the data are now encrypted. The same message is displayed on some of the other machines located in this and other areas of the business. Then he discovers that the company's file server has crashed, caused by the same problem: the WannaCryptor ransomworm. You are part of the incident response team:

a) Explain the incident response stages you might follow in order to mitigate the incident above. In your explanation refer to this particular incident. **[4 Marks]**

b) Discuss any **two** mechanisms or tools which a Network administrator could use to mitigate the spread of the worm and respond to this attack. **[4 Marks]**

c) Explain what a Unified threat management is and how it will benefit Fictitious Corp if implemented. **[4 Marks]**

d) Explain how each of the **four** layers of physical network security can help achieve network security physical network assets. **[8 Marks]**

Question 4

a. Explain **three** main purposes of carrying out backups. **[6 Marks]**

b. Discuss any **two** backup strategies that you can implement, in your discussion include the **one** advantage and disadvantage for each. **[6 Marks]**

c. Mention any two factors why it is important to test the Business Continuity Plan (BCP). **[2 Marks]**

d. Discuss the **three** network controls and give **one** example on each control. **[6 Marks]**

Question 5

a. Discuss how a firewall works and its benefits to an organization. **[4 Marks]**

b. Define a security bug then explain how it is a threat to our networks. **[4 Marks]**

c. Explain **two** types of password attacks. **[4 Marks]**

d. Explain the password rule of thumb. **[6 Marks]**

e. What is War dialing attack? **[2 Marks]**

END OF QUESTION PAPER